# WORKSHEET 1 : Multivariable Calculus (Limits, Continuity, Differentiability and Extrema)

**Exercise 1** *(**assigned to the students**)*

*Let $x = (x_1, \cdots, x_n) \in \mathbb{R}^n$. Define*

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|, \quad \|x\|_2 = \left(\sum_{i=1}^{n} |x_i|^2\right)^{1/2} \quad \text{and} \quad \|x\|_\infty = \max_{1\le i\le n} |x_i|.$$

1− *Show that $\|\cdot\|_2$ is a norm on $\mathbb{R}^n$.*

2− *Show that*

$$\|x\|_\infty \le \|x\|_2 \le \|x\|_1 \le n\|x\|_\infty \quad \text{and} \quad \|x\|_2 \le \sqrt{n}\|x\|_\infty,$$

*for all $x \in \mathbb{R}^n$.*

*Discuss the case $n = 1$.*

3− *Draw in $\mathbb{R}^2$ the closed unit ball*

$$\overline{B}(0,1) = \{(x,y) \in \mathbb{R}^2, \quad \|(x,y)\| \le 1\}$$

*using each of the norms $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$.*

**Exercise 2**

*In each case, determine and draw the domain of definition of the following functions :*

$$f(x,y) = \frac{\sqrt{-y+x^2}}{\sqrt{y}},\ f(x,y) = \frac{\ln(y)}{\sqrt{x-y}},\ f(x,y) = \frac{\sqrt{4-x^2-y^2}}{\sqrt{x^2+y^2-1}},\ f(x,y) = \ln(x-y^2).$$

**Exercise 3** *(**covered in class**)*

*Calculate the limit if it exists or show that it does not exist in the following cases :*

$$\lim_{(x,y)\to(1,1)} \frac{2}{x-y},\ \lim_{(x,y)\to(1,0)} \frac{y^3}{(x-1)^2+y^2},\ \lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2},$$

$$\lim_{(x,y)\to(0,0)} \frac{1+x^2+y^2}{y}\sin y,\ \lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}},\ \lim_{(x,y)\to(0,0)} \frac{xy}{x^2+xy+y^2},$$

$$\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{|\,x\,|+|\,y\,|},\ \lim_{(x,y)\to(0,0)} \frac{xy^4}{x^4+y^6},\ \lim_{(x,y)\to(0,0)} \frac{e^{xy}-1}{e^x-1}.$$

**Exercise 4**

*Let $\alpha, \beta > 0$. Study, depending on the values of $\alpha$ and $\beta$, the existence of a continuous extension of the function $f : \mathbb{R}^2 - \{(0,0)\} \to \mathbb{R}$, $(x,y) \mapsto f(x,y) = \dfrac{x^\alpha y^\beta}{x^2+y^2}$ at $(0,0)$.*

**Exercise 5**

*In each case, calculate the first partial derivatives of the following functions :*

$$f(x,y) = x^3 + 5xy^2 - 7y^7,\ f(x,y) = x^2\cos(e^{xy}),\ f(x,y,z) = x\sin(xz) - \ln(3-\sin^2(x+y))).$$

**Exercise 6**

*Let $f$ be the function defined on $\mathbb{R}^2$ by*

$$f(x,y) = \begin{cases} xy\left(\frac{x^2-y^2}{x^2+y^2}\right), & \text{if } (x,y) \neq (0,0), \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

1. *Show that $f$ is continuous on $\mathbb{R}^2$.*
2. *Compute $\nabla f(x,y)$.*
3. *Show that $f$ admits second partial derivatives at all points.*
4. *What can you deduce from the calculation of $\frac{\partial^2 f}{\partial x \partial y}(0,0)$, $\frac{\partial^2 f}{\partial y \partial x}(0,0)$, $\frac{\partial^2 f}{\partial x^2}(0,0)$ ?*

**Exercise 7**

*Study the continuity and $C^1$ class of the function $f$ defined by : $f(0,0) = 0$ and $f(x,y) = \dfrac{\sin(xy)}{|x|+|y|}$ if $(x,y) \neq (0,0)$.*

**Exercise 8** *(***covered in class***)*

*Define the function*

$$f(x,y) = \begin{cases} \frac{x^3-y^3}{x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{otherwise.} \end{cases}$$

*Show that $\frac{\partial f}{\partial x}(x,y)$ and $\frac{\partial f}{\partial y}(x,y)$ exist at every point in $\mathbb{R}^2$, that $f$ is continuous, but it is not differentiable at $(0,0)$.*

**Exercise 9**

*Let $f : \mathbb{R}^2 \to \mathbb{R}$,*

$$f(x,y) = \begin{cases} \frac{x^2y+y^2x}{x^2+y^2}, & \text{if } (x,y) \neq (0,0); \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

1. *Show that $f$ is continuous on $\mathbb{R}^2$.*
2. *Show that $f$ is not differentiable at $(0,0)$.*

**Exercise 10**

*Let $f : \mathbb{R}^2 \longmapsto \mathbb{R}^2$ be defined by*

$$f(x,y) = \begin{cases} \left(\dfrac{xy}{(x^2+y^2)^\alpha}, xy\right), & \text{if } (x,y) \neq (0,0); \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

*Determine the values of $\alpha$ for which $f$ is continuous, differentiable, and of class $C^1$.*

**Exercise 11**

*Let $f$ be the real function defined on $\mathbb{R}^2$ by $f(x,y) = \sin(x^2 - y^2)$. Let $g$ be the function from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by $g(x,y) = (x+y, x-y)$.*

1. *Compute the partial derivatives of $f \circ g$ and the differential of $f \circ g$ at the point $(x,y)$.*
2. *Compute the Jacobian matrices of $f$ and $g$ at $(x,y)$.*
3. *By applying the theorem on the composition of differentiable functions, find the Jacobian matrix of $f \circ g$ at $(x,y)$.*

**Exercise 12** *(**assigned to the students**)*

*Let the function be defined by*

$$f(x,y) = (x^2 + x\cos y, e^{x-y}, y^3 x).$$

1. *Write the Jacobian matrix $Jf$ at any point in $\mathbb{R}^2$.*
2. *Study the differentiability of $f$ on $\mathbb{R}^2$.*
3. *Determine $Df(a)$, the differential of $f$ at point $a \in \mathbb{R}^2$.*

**Exercise 13** *(**assigned to the students**)*

*Let $g$ be the mapping from $\mathbb{R}^2$ to itself defined by*

$$g(u,v) = (u^2 + v^2, u^2 - v^2).$$

*Compute the Jacobian matrix of the mapping $h : \mathbb{R}^2 \to \mathbb{R}^2$ defined by*

$$h(x,y) = g(x+y, xy).$$

**Exercise 14**

*I) Compute $z'(t)$ or $w'(t)$ :*

1− $z(t) = f(x(t), y(t)),\ f(x,y) = x^2 + y^2 + xy,\ x(t) = \sin(t),\ y(t) = e^t.$

2− $z(t) = f(x(t), y(t)),\ f(x,y) = \cos(x+y),\ x(t) = t^3,\ y(t) = 4.$

3− $w(t) = f(x(t), y(t), z(t)),\ f(x,y,z) = xe^{z/y},\ x(t) = t,\ y(t) = 4 - t,\ z(t) = 2t.$

4− $w(t) = f(x(t), y(t), z(t)),\ f(x,y,z) = \log(\sqrt{x^2 + y^2 + z^2}),\ x(t) = \sin(t),\ y(t) = \cos(t),\ z(t) = \tan(t).$

*II) Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be a $\mathcal{C}^2$ function and $g : \mathbb{R}^2 \longrightarrow \mathbb{R}$ defined by*

$$g(u,v) = f(u+v, u-v)$$

1− *Justify that $g$ is differentiable.*

2− *Express $\dfrac{\partial g}{\partial u}$ and $\dfrac{\partial g}{\partial v}$ in terms of the partial derivatives of $f$ denoted $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.*

3− *Justify that $g$ is of class $\mathcal{C}^2$.*

4− *Express the second-order partial derivatives of $g$ in terms of the partial derivatives of $f$.*

**Exercise 15**

*For each of the following functions, study the nature of the given critical point :*

1. $f(x,y) = x^2 - xy + y^2 + 1$ *at critical point* $(0,0)$ *;*
2. $f(x,y) = x^2 + 2xy + y^2 + 10$ *at critical point* $(0,0)$ *;*
3. $f(x,y) = x^3 + 2xy^2 - y^4 + x^2 + 3xy + y^2 + 10$ *at critical point* $(0,0)$ *;*

**Exercise 16**

*Find the critical points of the following functions and determine whether they are local minima, local maxima, or saddle points.*

1. $f(x,y) = x^3 + 6x^2 + 3y^2 - 12xy + 9x$ *;*
2. $f(x,y) = \sin x + y^2 - 2y + 1$ *;*

3. $f(x,y) = \exp(xy)(xy-1) + y(y-4)$ ;
4. $f(x,y,z) = \dfrac{x^2}{2} + xyz - z + y$ ;
5. $f(x,y,z) = (x+y+z)^2$ ;

**Exercise 17**

*Let $f$ be the mapping defined on $\mathbb{R}^2$ by $f(x,y) = (x^4+y^4) - (x-y)^2$. Determine the critical points of $f$ and establish their nature.*

**Exercise 18**

*Use the method of Lagrange multipliers to calculate the maximum and minimum of the function $f$ under the indicated constraint :*

1. $f(x,y) = 3x + y$ *under the constraint* $x^2 + y^2 = 10$.
2. $f(x,y) = -x^2 + y^2$ *under the constraint* $x^2 + 4y^2 = 4$.
3. $f(x,y) = xy$ *under the constraint* $x^2 + y^2 = 1$.